C1.

1.Parameter (μ): The true mean average shopping cadence (days between orders) for the entire population of Instacart customers.

2. **Appropriate Test: Z-test**

Justification:

In big data analytics when the population parameter is computationally accessible (as it is here), using a Z-test provides a more precise exact test compared to the t-test, which is traditionally an approximation for smaller samples with unknown variances.

Known Population Parameters: Z-distribution is used when the population standard deviation is known. In this assignment, we calculated the exact standard deviation from the entire Instacart dataset (~200k users) which came out to be 6.915575. So, we have all parameters.

Sample Size: The Central Limit Theorem states that a Z-distribution is appropriate for large samples where n is greater than or equal to 30. For this dataset a sample size of 100 was chosen, in this way the sampling distribution of the mean is guaranteed to be normal, making the Z-score a precise choice.

3.

Hypotheses:H_0: μ=7 (Weekly cycle) vs H_a: μ≠7(non-weekly cycle)

Sample mean: 16.1417

Hypothesized Mean: 7

Standard error (SE): $\frac{\sigma}{\sqrt{n}} = \frac{6.9156}{\sqrt{100}} = 0.6916$

$Z = \frac{sample\ mean - \mu_0}{SE} = \frac{16.1417-7}{0.6916} = \frac{9.1417}{0.6916} \approx 13.22$

4. verification using Python

TEST STATISTIC

#Verifying test statistic

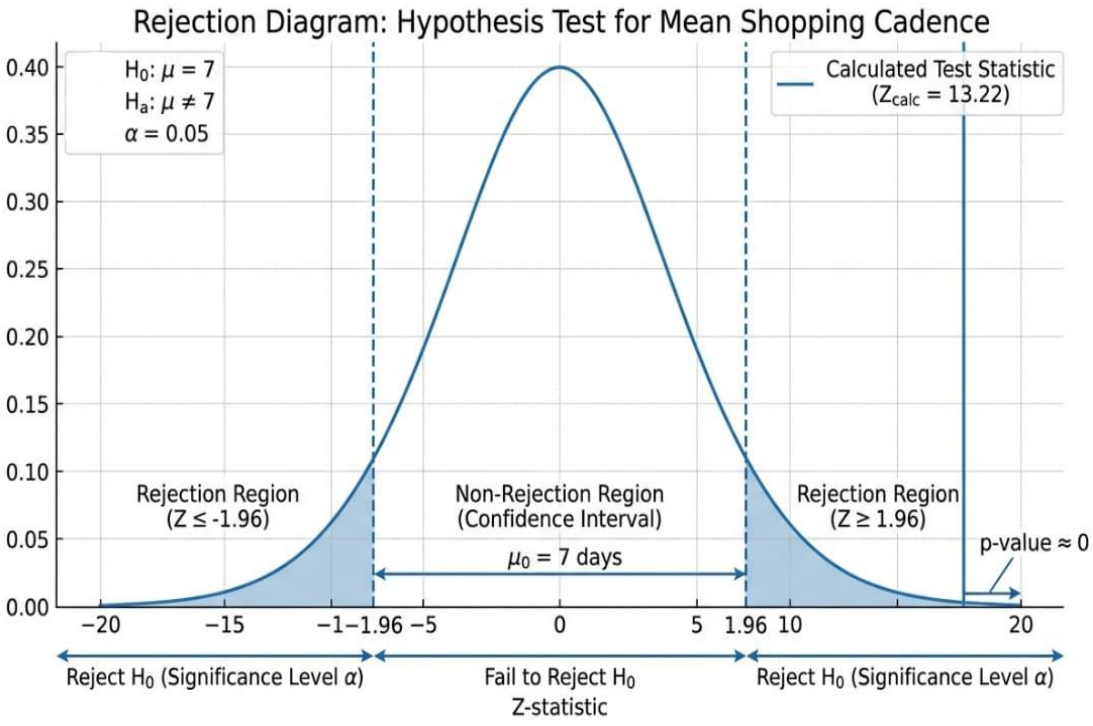
```
sample_mean = 16.1417
pop_sd = 6.915575
n = 100
hyp_mean = 7

# Calculate Z-stat and p-value
z_stat = (sample_mean - hyp_mean) / (pop_sd / np.sqrt(n))
p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))

print(f"Z-statistic: {z_stat:.2f}")
print(f"p-value: {p_value:.4e}")
```

```
Z-statistic: 13.22
p-value: 0.0000e+00
```

5.



6. Conclusion:

Since the calculated test statistic (13.22) is significantly greater than the critical value (1.96), and the p-value is essentially zero, we **reject the null hypothesis**.

There is overwhelming statistical evidence to conclude that the true mean shopping cadence is not 7 days.

7.Business Interpretation

- Finding: Customer reorder behaviour differs drastically from a strict weekly cycle. The point estimate of 16.14 days suggests that the average customer takes more than twice as long as a week to return to the platform.

- Strategic Advice: Management should stop planning logistics around a "universal weekly shop" and instead diversify fulfillment schedules to account for this longer 16-day average return window. Marketing should target users around the 14-day mark with promotions to align with their actual behavioural rhythm rather than forcing a weekly schedule.

C2.

1.Point estimates/ GROUP MEANS

```
# 1. Compute order sizes
order_size = orderprodprior.groupby('order_id').size().reset_index(name='order_size')

# 2. Merge 'order_size' into 'orders' DataFrame
# check if 'order_size' is already there to prevent KeyError on re-runs
if 'order_size' not in orders.columns:
    orders = orders.merge(order_size, on='order_id')

# 3. Create the Weekend Indicator
# Sunday (0) and Saturday (6) are marked as 1, others as 0
orders['weekend'] = orders['order_dow'].apply(lambda x: 1 if x in [0, 6] else 0)

# 4. Compute Group Means (Point Estimates)
group_means = orders.groupby('weekend')['order_size'].mean()

# Assign to specific variables for formal hypothesis testing
x_bar_w = group_means[1]  # Mean Weekend Order Size (weekend is 1)
x_bar_d = group_means[0]  # Mean Weekday Order Size (weekday is 0)

print(f"Point Estimate Weekend (x_bar_w): {x_bar_w:.4f} items")
print(f"Point Estimate Weekday (x_bar_d): {x_bar_d:.4f} items")
```

```
Point Estimate Weekend (x_bar_w): 10.9664 items
Point Estimate Weekday (x_bar_d): 9.7060 items
```

- **Mean Weekend Order Size (x_w):** 10.9664 items
- **Mean Weekday Order Size (x_d):** 9.7060 items
- **Initial Observation:** In the sample, weekend orders are larger than weekday orders by approximately **1.26 items.**

2.

Null Hypothesis (H_0): $\mu_{weekend} = \mu_{weekday}$

- *(There is no difference in the true population mean order size between weekends and weekdays)*

Alternative Hypothesis (Ha): $\mu_{weekend} \neq \mu_{weekday}$

- *(The true population mean order size on weekends differs significantly from weekdays)*

3. Justify independent samples t-test

Comparison of Two Means: A numeric variable (order size) is being measured across two distinct and unrelated populations (Weekend vs. Weekday shoppers).

Independence: The groups are unrelated, a single transaction belongs to either a weekend or a weekday, but not both.

Overall, the independent t-test is most ideal because the high-volume Sunday rush and quieter mid-week operations likely result in unequal variances.

4. Computing Test Statistic

The test calculates the difference between group means divided by the estimated standard error of the difference.

```
# 1. Extract the specific groups for the test
# Using the 'orders' DataFrame and 'weekend' indicator
weekend_basket_sizes = orders[orders['weekend'] == 1]['order_size']
weekday_basket_sizes = orders[orders['weekend'] == 0]['order_size']

# 2. Compute the Test Statistic
# We use Welch's t-test (equal_var=False)
t_stat, p_val = stats.ttest_ind(weekend_basket_sizes, weekday_basket_sizes, equal_var=False)

print(f"Calculated t-statistic: {t_stat:.4f}")
print(f"Calculated p-value: {p_val:.4e}")

# 4. Decision Rule Check (Alpha = 0.05)
if p_val < 0.05:
    print("Conclusion: Reject the Null Hypothesis (H0).")
else:
    print("Conclusion: Fail to Reject the Null Hypothesis (H0).")
```

```
Calculated t-statistic: 136.2849
Calculated p-value: 0.0000e+00
Conclusion: Reject the Null Hypothesis (H0).
```

t-statistic= 136.2849

p-value= 0

- This value represents the probability of observing a difference of 1.26 items if there were truly no difference between weekend and weekday populations.

5.

A p-value of 0 means there is a **0% probability** of observing the sample results assuming the Null hypothesis is true.

Decision Rule: If calculated p-value < 0.05, reject the null hypothesis.

Statistical Conclusion: There is sufficient statistical evidence to conclude that the true population mean order size for weekends differs significantly from that of weekdays.

Interpretation: Given the massive volume of the Instacart dataset and the observed point estimates (**10.97** items vs **9.71** items) the p-value is near zero ($p<0.001$), providing strong to overwhelming support for the alternative hypothesis.

6.

- **The Magnitude:** Weekend orders contain approximately **13% more items** than weekday orders (10.97 vs 9.71 items).
- **Operational Impact:** A difference of **1.26 items per order** is highly significant at scale. For a fulfillment center processing tens of thousands of orders, this increase in basket size leads to:
    1. **Increased Labor:** Significantly higher picking and packing time per order on Saturdays and Sundays.
    2. **Logistics Bottlenecks:** Greater demand for storage space in delivery vehicles and potential delays in fulfillment.

Strategic Recommendation: Management should prioritize resources on **Sundays and Mondays** (the peak demand windows) and optimize warehouse routing to handle the larger, more complex weekend baskets.

C3.

1.

```
# group the items by order_id and count them to define 'order_size'
order_items = pd.read_csv("../content/drive/MyDrive/ADA2026_Instacart_Project/data/raw/instacart/order_products__prior.csv")
order_size_df = order_items.groupby('order_id').size().reset_index(name='order_size')

# Merge 'order_size' into main 'orders' DataFrame
if 'order_size' not in orders.columns:
    orders = orders.merge(order_size_df, on='order_id')

# Handle Missing Values
# First-time orders have NaN in 'days_since_prior_order' and must be removed for correlation
clean_data = orders[['days_since_prior_order', 'order_size']].dropna()

# Random Sampling for Visual Clarity (n=30,000)
# This prevents 'Overplotting Density' caused by the massive dataset size
tmp_sampled = clean_data.sample(n=30000, random_state=42)

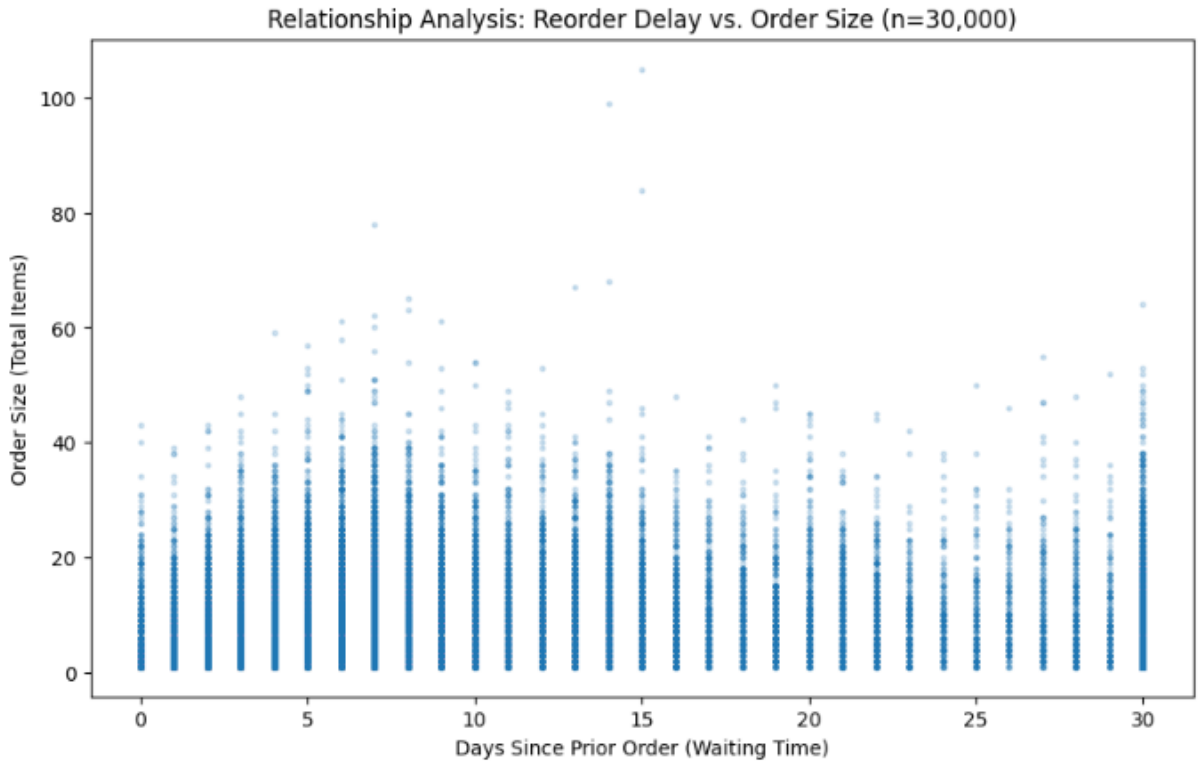
# Construct the Scatter Plot
plt.figure(figsize=(10, 6))

# Use small markers (s=5) and low alpha (0.2) to reveal density patterns
plt.scatter(tmp_sampled['days_since_prior_order'],
            tmp_sampled['order_size'],
            alpha=0.2,
            s=5,
            color='tab:blue')

# Formatting and Labels

plt.xlabel("Days Since Prior Order (Waiting Time)")
plt.ylabel("Order Size (Total Items)")
plt.title("Relationship Analysis: Reorder Delay vs. Order Size (n=30,000)")
plt.show()
```

Result:



Direction: Based on the dense clustering along the bottom of the Y-axis, there is an upward trend (positive relationship), suggest a "stockpiling" effect where longer delays lead to more items.

The 30-Day Spike: there is a prominent vertical column at the 30-day mark. This is the "capped" limit of the Instacart data, which represents a significant behavioural segment that may influence the regression slope.

2.

Correlation formula:

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

To extract these values from the scatter plot:

```
# clean_data was defined in previous steps as orders[['days_since_prior_order', 'order_size']].dropna()
x = clean_data['days_since_prior_order']
y = clean_data['order_size']

# 2. Calculate the six necessary summary components
n = len(clean_data)
sum_x = x.sum()
sum_y = y.sum()
sum_x2 = (x**2).sum()
sum_y2 = (y**2).sum()
sum_xy = (x * y).sum()

# 3. Print the values
print(f"n: {n}")
print(f"Sum of X (sum_x): {sum_x}")
print(f"Sum of Y (sum_y): {sum_y}")
print(f"Sum of X^2 (sum_x2): {sum_x2}")
print(f"Sum of Y^2 (sum_y2): {sum_y2}")
print(f"Sum of XY (sum_xy): {sum_xy}")
```

```
n: 3008665
Sum of X (sum_x): 32214505.0
Sum of Y (sum_y): 30356421
Sum of X^2 (sum_x2): 586155195.0
Sum of Y^2 (sum_y2): 476847725
Sum of XY (sum_xy): 337079953.0
```

Computation:

```
# Values from Above
n = 3008665
sum_x = 32214505.0
sum_y = 30356421
sum_x2 = 586155195.0
sum_y2 = 476847725
sum_xy = 337079953.0

# Formula: r = (n*sum_xy - sum_x*sum_y) / sqrt([n*sum_x2 - sum_x^2] * [n*sum_y2 - sum_y^2])
numerator = (n * sum_xy) - (sum_x * sum_y)
denominator = np.sqrt((n * sum_x2 - sum_x**2) * (n * sum_y2 - sum_y**2))
r = numerator / denominator

print(f"Correlation Coefficient (r): {r:.4f}")
```

```
Correlation Coefficient (r): 0.0594
```

$$r = \frac{(3{,}008{,}665 \; x \; 337{,}079{,}953) - (32{,}214{,}505 \; x \; 30{,}356{,}421)}{\sqrt{[(3{,}008{,}665 \; x \; 586{,}155{,}195) - (32{,}214{,}505^{2)} \; x \; (3{,}008{,}665 \; x \; 476{,}847{,}725) - (30{,}356{,}421^{2})]}}$$

$$r = \frac{36{,}243{,}743{,}215{,}640}{\sqrt{725{,}766{,}981{,}397{,}650 \; x \; 513{,}163{,}252{,}779{,}884}} = 0.0594$$

Interpretation: An r value of **0.0594** indicates a **very weak positive linear relationship**, this means that as customers wait longer to reorder, there is a very slight (almost negligible) tendency for their order size to increase, suggesting waiting time is a poor predictor of order size.

3.

Test if this correlation is genuine or due to chance using the -test for significance.

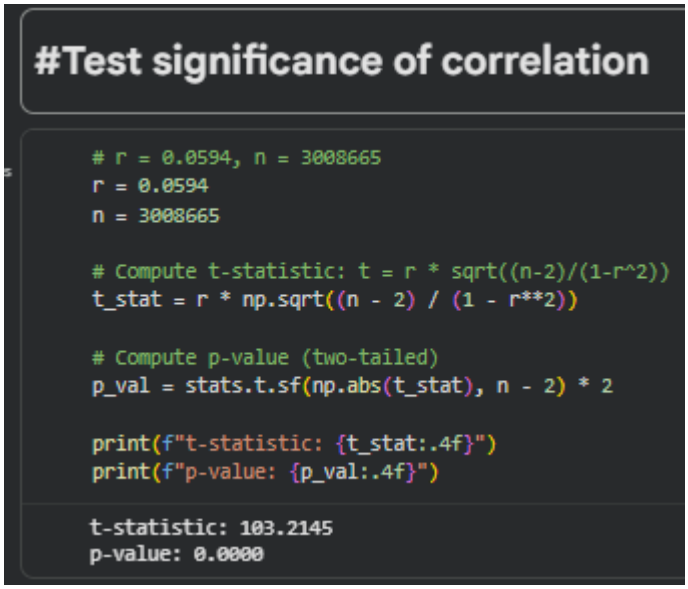
Hypotheses: H_0: p = 0

H_a: p ≠ 0

Test statistic formula: $t = r\sqrt{\frac{n-2}{1-r^2}}$

$$\frac{0.0594}{\sqrt{\frac{3{,}008{,}663}{1-0.0594^2}}} = 103.25$$

Validation:



Decision: With n>3million, the critical value for a two-tailed test is 1.96. Since 103.25> 1.96, we reject H_0.

Conclusion: The relationship is statistically significant, but only because the massive sample size makes even a tiny correlation mathematically detectable.

4.

Using the Ordinary Least Squares method to estimate the line y=b_0 + b_1x

$b_1 = \frac{36{,}243{,}743{,}215{,}640}{725{,}766{,}981{,}397{,}650} = 0.0499$

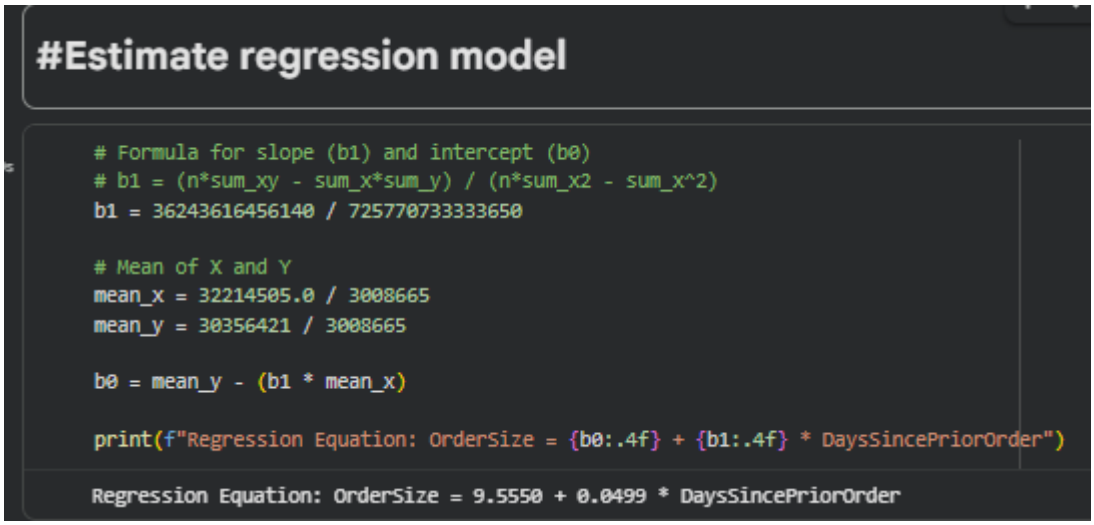
Mean of X: $\frac{32,214,505}{3,008,665}$ = 10.707

Mean of Y: $\frac{30,356,421}{3,008,665}$= 10.090

Intercept (b_0): y-b_1x= 10.090- (0.0499 x 10.707) = 9.555

ESTIMATED REGRESSION MODEL: OrderSize = 9.555 + 0.050(DaysSincePriorOrder)

Verification:



5.

Intercept ($b_0 = 9.555$): A customer returning immediately (0 days delay) is predicted to buy approximately 9.56 items.

Slope (b_1=0.050): This is the marginal rate of change. For every 1 additional day a customer waits to reorder, their basket size is predicted to increase by only 0.05 items.

MANAGERIAL INSIGHT: It would take 20 days of waiting for a customer to add just one extra item to their order. This suggests that stockpiling behaviour is negligible.

6.

$R^2 = r^2 = 0.0594^2 = 0.0035$

Interpretation: Only 0.35% of the variation in order size is explained by the number of days since the prior order. The remaining 99.65% is determined by factors external to the waiting time, such as household size, dietary needs, or marketing promotions.

7.

1. Linearity: The model assumes a straight-line relationship. However, the extremely low suggests that a linear model may be an oversimplification and that order size is largely independent of time.
2. Homoscedasticity: It is assumed that the variability of order sizes remains constant across all waiting times. If users waiting 30 days have wildly different basket sizes compared to 7-day users (a "fan" shape in the residuals), this assumption is violated, making the model less reliable for those segments